

Wenhan Zhang

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EDUCATION

Santa Clara University, Santa Clara, CA

Expected Jun 2028

B.S. in Computer Science and Engineering, Machine Learning Concentration | GPA: **3.6**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Linear Algebra, Probability & Statistics, Calculus

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript

AI / ML: PyTorch, LLM Evaluation, RAG, FAISS, Whisper, OCR, OpenAI API

Frameworks & Systems: FastAPI, SQLite/FTS5, MCP, Docker, Git, Electron, Node.js, Firebase, macOS, Linux

EXPERIENCE

Baidu

Summer 2026

Machine Learning Engineer Intern | **Python, LLM Agents, Evaluation**

- Designed **repeatable evaluation workflows** for LLM agents across reasoning, instruction following, and tool use, standardizing benchmark execution and result analysis.
- Curated **benchmark prompts and evaluation datasets**; automated batch execution, result aggregation, and report generation in **Python**.
- Performed **regression testing across model versions** and analyzed failure patterns, producing structured reports to support iterative debugging.

Resilience Inc.

Summer 2025

Software Developer Intern | **Electron, Node.js, Python, Firebase, OpenAI API**

- Built a **cross-platform desktop event-capture service** in Electron, Node.js, and Python to collect and process application events.
- Integrated **Firebase and OpenAI APIs** into the event pipeline for cloud persistence and AI-assisted processing.
- Designed modular interfaces across **collection, storage, and inference components** to support maintainable extension of the system.

Santa Clara University

Sep 2025 - Present

Computer Science Teaching Assistant / Grader / Tutor

- Supported students in **programming fundamentals and data structures** through tutoring and **debugging guidance**, strengthening problem-solving and software development skills.

PROJECTS

AI Desk Presence - Agent Handoff Runtime | **Python, ESP32, SQLite, A2A, macOS** [[GitHub](#)]

Jul 2026

- Built an **event-driven Python runtime** with a **four-state work-session machine**, SQLite persistence, daily statistics, stable event payloads, and fault-isolated listeners behind one public API.
- Integrated macOS context capture with a **reconnecting ESP32/mmWave USB-serial adapter**, decoupling physical-presence sensing from session, break, and timeline services.
- Implemented a **presence-aware A2A 1.0 orchestrator** that claims an armed task **exactly once** on absence and returns a macOS notification, **report.md**, and **artifact.json** when the user returns.

AI Agent Harness - Agent Runtime & Evaluation | **Python, FastAPI, MCP, RAG** [[GitHub](#)]

Jun 2026

- Built a **multi-step LLM agent runtime** with schema-based tools, permission enforcement, short-term memory, context retrieval, and **9 provider adapters**.
- Implemented **stdio MCP integration**, **cited local RAG**, **replayable traces**, **FastAPI/Docker deployment**, and a **static dashboard** for debugging and comparing agent runs.
- Developed **JSONL evaluation and provider leaderboards** measuring task success, tool accuracy, latency, token use, cost, determinism, and failure categories.

MindTrace - Multimodal AI Memory | **Python, Whisper, OCR, FTS5, FAISS** [[GitHub](#)]

Apr 2026

- Architected a **local-first desktop memory assistant** that captures audio and screen content through Whisper and OCR into portable JSON sessions and searchable SQLite history.
- Designed **hybrid retrieval with SQLite FTS5 and FAISS/numpy** over **300-600-token chunks**, followed by deduplication, reranking, and token-budgeted context packing.
- Built **durable background vector indexing and pre-OCR frame deduplication**, with sensitive-app exclusion and Markdown/PDF export for privacy-aware desktop workflows.

LEADERSHIP

President, Santa Clara University Billiards Club

Sep 2025 - Present

- Planned and organized tournaments introducing students to **snooker, American 8-ball, and American 9-ball**.